

Li Zhang

✉ lilyzhang4751@gmail.com | 🌐 github.com/lilyzhang4751 | 🔗 linkedin.com/in/lily-zhang-97b187228
Willing to relocate to Germany | Chinese (Native) | English (Professional) | German (Currently learning)

Java Backend Developer with 4.5 years of experience building high-traffic logistics backend systems at Pinduoduo. Strong background in Java, Spring Boot, Kafka, MySQL, Redis, RESTful APIs, asynchronous processing, and production troubleshooting. Supported backend services where peak traffic reached around 5,000 QPS and over 100 million daily operation requests during promotional peaks. Strong focus on concurrency control, idempotency, transaction boundaries, and Kafka-based asynchronous workflows.

SKILLS

Backend Development: Java, Spring Boot, RESTful APIs, Dubbo/RPC, MyBatis, Spring Data JPA/Hibernate

Data / Messaging: MySQL, PostgreSQL, Redis, Elasticsearch, Kafka, SQL, transactions, indexing, async processing, idempotency

Engineering / Ops: Maven, Git, Docker, Linux, Kubernetes, CI/CD, Kibana, canary deployments

Project Stack: Spring Security, JWT, JUnit 5, Testcontainers, GitHub Actions

EXPERIENCE

Pinduoduo

Dec 2020 - Jul 2025

Java Backend Developer

Tech Stack: Java, Spring Boot, Dubbo RPC, Kafka, MySQL, MyBatis, Redis, Elasticsearch, Kubernetes

Worked on Duoduo Parcel Pickup Station, a high-traffic backend platform used by pickup station operators to manage parcel lifecycle data and trigger downstream notifications.

- **Backend:** Developed Java/Spring Boot/Dubbo RPC backend services for core parcel workflows, including preprocessing, check-in/check-out operations, inventory checks, shelf changes, returns, and status transitions.
- **Scale:** Supported backend APIs where preprocessing traffic reached around 5,000 QPS and daily core operation requests could exceed 100 million during traffic peaks.
- **Concurrency & Consistency:** Used Redis-based distributed locking, Redis caching, MySQL sharding, and local transactions to support high-concurrency parcel operations and maintain consistency between different records.
- **Messaging & Reliability:** Implemented Kafka-based asynchronous notification workflows, including intelligent voice, delayed SMS delivery, cancellation handling, retry handling, and idempotent processing.
- **Search & Observability:** Used Elasticsearch for cross-station and complex parcel searches, and used Kibana and internal metrics platforms for log analysis, monitoring, production troubleshooting, and release support in a Kubernetes-based environment.

Guangzhou Huizhi Communication Technology Co., Ltd.

Jul 2017 - Jun 2019

Pre-sales Engineer

- **Summary:** Supported telecommunications projects through technical solution support, bidding documentation, and customer communication, building strong requirement analysis and cross-functional collaboration skills.

PROFESSIONAL DEVELOPMENT

Career Break & Backend Engineering Refresh

Aug 2025 - Present

Took a planned career break after leaving Pinduoduo, including travel, relocation preparation, and focused backend engineering refresh for the German job market.

- **Project:** Built Parcel-Hub-Core, a Java 21 / Spring Boot logistics backend project covering parcel inbound/outbound workflows, user authentication, parcel status transitions, and event publishing.
- **Engineering Focus:** Practiced Kafka-based asynchronous messaging, Transactional Outbox, PostgreSQL, SpringData JPA/Hibernate, Redis, Spring Security JWT, integration testing with Testcontainers, and CI workflows with GitHub Actions.

EDUCATION

University of Bristol

Sep 2019 - Dec 2020

MSc in Computer Science

Relevant coursework: Object-Oriented Programming with Java, Software Engineering, Databases, Computer Architecture.

Jilin University

Aug 2013 - Jun 2017

BEng in Communication Engineering

Engineering foundation in communication systems, computer fundamentals, mathematics, and problem-solving.